

# Comparison of CNN and QCNN Performance in Binary Classification of Breast Cancer Histopathological Images

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**Abstract**—Breast cancer is a leading cause of mortality among women, with early detection being crucial for improving outcomes. This study explores the use of Quantum Convolutional Neural Network (QCNN) for automating breast cancer detection through histopathological images. We compare QCNNs with classical Convolutional Neural Network (CNN) for binary classification of benign and malignant tumors using the BreakHis dataset. Our results demonstrate that QCNN outperform CNN in accuracy, especially in smaller batch sizes, and offer faster convergence with improved computational efficiency. This highlights QCNNs' potential in enhancing diagnostic precision and efficiency in medical imaging.

**Index Terms**—Breast cancer, QCNN, CNN, histopathological images, quantum computing.

## I. INTRODUCTION

Breast cancer remains one of the most prevalent and deadly malignancies among women worldwide, making it crucial to develop better ways to detect it early and improve treatment options. Breast Cancer Disease (BCD) involves the abnormal and unchecked proliferation of breast tissue. The World Health Organization (WHO) identifies BCD as the second most common cause of cancer mortality in women. Fortunately, the National Breast Cancer Foundation (NBCF) highlights that early detection significantly improves the chances of successful treatment. Therefore, advanced systems are needed to detect BCD at its earliest stages [1].

Invasive Ductal Carcinoma (IDC), or infiltrating ductal carcinoma, is the most common type of breast cancer, comprising about 80% of cases [2]. It starts in the milk ducts and can spread to nearby breast tissue, lymph nodes, and other parts of the body. The 5-year survival rate for breast cancer is approximately 89%. Automating the detection of IDC could greatly aid pathologists by streamlining the diagnostic process and reducing errors.

Detecting IDC is a complex and time-intensive task, typically requiring pathologists to examine histopathological samples under a microscope visually. Automating the detection of this cancer type could significantly benefit pathologists by accelerating the diagnostic process and minimizing potential errors.

In recent years, the automation of medical imaging has garnered significant attention in research. Techniques like

the Scalar Invariant Fourier Transform (SIFT), Speeded-up Robust Features (SURF), and Local Binary Pattern (LBP) have been employed to classify breast histopathological images as either benign or malignant [3], [4], [5]. Convolutional Neural Networks (CNNs), which have become the cornerstone of image recognition tasks due to their layered architecture that effectively extracts and identifies relevant features, are utilized to classify histopathological images as benign or malignant [6]. These layers collaborate to progressively refine image features, allowing the network to accurately recognize and classify images.

The decision to utilize CNN stems from their established success in medical image analysis due to their ability to learn hierarchical features directly from image data. Similarly, QCNN was selected to explore the potential of quantum computing in overcoming the computational limitations of traditional CNN. This research aims to determine whether QCNN can overcome the limitations of CNN, providing more accurate breast cancer diagnosis and making it suitable for resource-intensive applications like medical imaging.

As input data size or model depth increases, CNN face growing computational and resource demands, underscoring the need for new models or learning environments to address these challenges. Quantum computing, leveraging the principles of superposition, entanglement, and parallelism, introduces algorithms that far exceed the capabilities of classical computers. These quantum algorithms provide innovative solutions to complex problems that classical computing struggles with due to limitations in time and resources [6]. Quantum computers use quantum bits (qubits) to encode and process information, allowing them to simultaneously exist in a superposition of  $|0\rangle$  and  $|1\rangle$  [7]. In deep learning, quantum circuits with flexible gates can assist in backpropagation by modifying gate parameters and carrying out measurements, thereby refining the learning process.

The motivation for this research lies in the potential of quantum computing, especially in quantum machine learning (QML). QCNN offer a novel solution to the limitations of classical CNN, leveraging the principles of superposition and entanglement to enhance both computational efficiency and diagnostic accuracy. QCNN have the potential to revolutionize

fields like medical imaging and autonomous driving. In this study, we apply QCNN to the challenging task of breast cancer image classification, where histopathological images exhibit wide inter-class variation and complex geometrical shapes. Using a histopathological database that includes benign and malignant tumor images, we compare the performance of CNN and QCNN in binary classification, focusing on key factors such as training accuracy, batch size, and the number of epochs. The results highlight QCNNs' superior performance and their potential to advance medical imaging technology.

The key contributions of our research are as follows:

- We created a detailed experimental framework that involved data collection, preprocessing, and dataset preparation tailored for binary classification tasks.
- We developed and optimized CNN and QCNN models, with a focus on tuning critical parameters like batch size and epoch count.
- We carried out a comprehensive performance comparison, employing standard evaluation metrics to assess the CNN and QCNN models.

The structure of this paper is organized as follows: Section II reviews the existing literature. In Section III, we compare the architectures of CNN and QCNN models. Section IV details our methodology, including data sourcing, tools, training environment, and hardware and software integration. Section V discusses the evaluation metrics used to assess model performance. Section VI outlines the experimental setup, detailing the configurations and conditions employed to train and evaluate both CNN and QCNN models. Section VII offers an in-depth analysis of our experimental findings. Finally, Section VIII concludes the paper by summarizing the key insights and proposing directions for future research.

## II. LITERATURE REVIEW

The application of CNN and QCNN to breast cancer histopathological image classification has emerged as a crucial research area in medical image analysis. CNN, a deep learning method designed to process structured grid data such as images, have been widely used in breast cancer classification due to their ability to automatically learn and extract hierarchical features. QCNN, a novel extension of CNN leveraging quantum computation, show potential in improving the efficiency and accuracy of traditional CNN by utilizing quantum principles such as superposition and entanglement. QCNN offer potential advantages in handling high-dimensional datasets by processing image features in parallel, reducing computational complexity described by Meedinti et al. (2023) [8]. Despite their promising performance in areas like medical imaging, QCNN face challenges, including limited quantum hardware, inefficient training algorithms, and scalability issues noted by Rajesh et al. (2021) [9].

Teresa Araújo et al. (2017) introduced a CNN-based method to classify breast cancer histology images into normal, benign, in situ, and invasive carcinoma categories, achieving an overall accuracy of 77.8% [5]. This study demonstrated that CNN are effective in learning multi-scale features from histopathology

images, allowing them to identify carcinoma cases with high sensitivity (95.6%). However, as CNN are computationally expensive and rely heavily on large datasets for high accuracy, their performance can sometimes be hindered by the availability of annotated data, especially for complex medical imaging tasks.

The study by Spanhol et al. (2016) compared the performance of CNN and QCNN in binary classification of breast cancer histology images [10]. The authors highlighted the advantages of using CNN over traditional machine learning models that rely on hand-crafted features, noting that CNN outperformed previous models in recognizing cancerous tissue patches. However, while CNN achieved promising results, the study emphasized the potential of QCNN to surpass traditional CNN in terms of efficiency, as quantum algorithms could process image features in parallel, reducing the computational complexity associated with deep learning. Despite their potential, QCNN face challenges such as limited quantum hardware, inefficient training algorithms, and scalability issues.

In contrast, the introduction of QCNN addresses some of these computational limitations. While the literature on QCNN remains relatively sparse compared to CNN, their potential is highlighted in works like Spanhol et al. (2016), where QCNN showed promising preliminary results in reducing the computational burden associated with deep learning by leveraging quantum mechanics [10]. By processing features through quantum states, QCNN may offer faster learning rates and improved model generalization, particularly in scenarios involving large, complex datasets like breast cancer histology images.

Further showcasing its versatility, Muhammad Al-Zafar Khan et al. demonstrated the application of QCNN in brain tumor classification, achieving an impressive accuracy of 99.67% [11]. This highlights QCNN's potential not only in breast cancer diagnosis but also in broader clinical applications, where rapid and reliable diagnoses are critical. The study underscores QCNN's ability to facilitate precise classification in medical imaging, reinforcing the need for continued exploration of quantum methodologies in healthcare.

## III. COMPARISON OF CNN AND QCNN ARCHITECTURES

### A. CNN Architecture Overview

We explored the architecture of a CNN as an effective framework for image classification, as depicted in Fig.1. CNN are a widely used approach for analyzing histopathological images of breast cancer. The process begins with raw breast tissue images, which are passed through convolutional layers. These layers use filters to identify important spatial features, such as edges and textures, that help distinguish between benign and malignant cases. As the data flows through multiple layers, the network progressively learns more complex and abstract patterns, enabling accurate feature extraction.

After the convolutional layers, pooling layers are applied to simplify the feature maps by reducing their dimensions. Techniques like max pooling or average pooling retain only the most significant features, reducing computational complexity

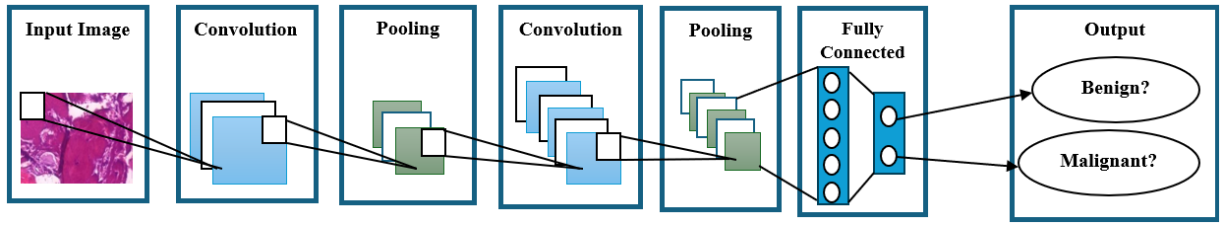


Fig. 1. Diagram of CNN Architecture.

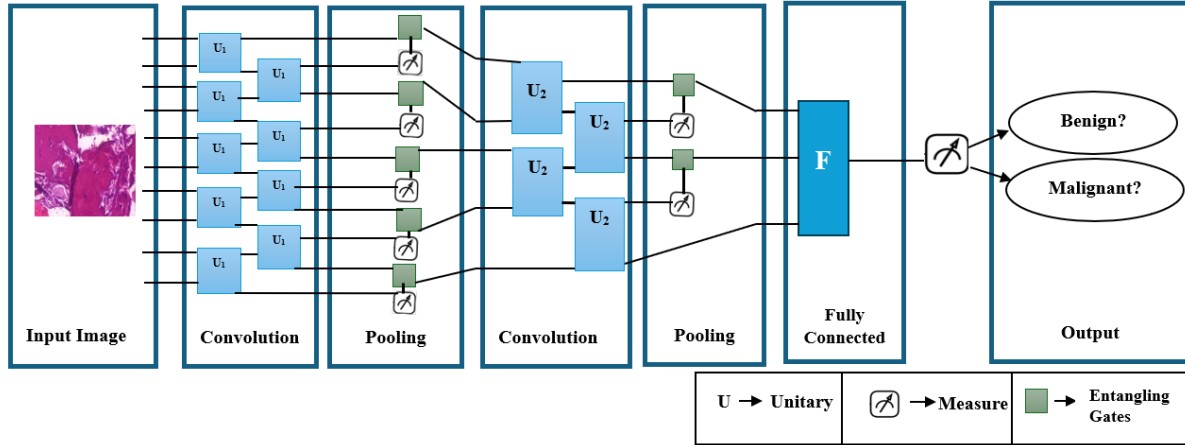


Fig. 2. Diagram of QCNN Architecture.

while preserving critical information. The simplified features are then passed to a fully connected layer, where they are combined into a compact representation. Finally, the network predicts whether the input image represents a benign or malignant case. This architecture is particularly adept at handling the complexities and variations found in breast cancer images, making it a powerful tool for automated diagnosis and playing a crucial role in the early detection and treatment of breast cancer [7].

#### B. QCNN Architecture Overview

QCNN extend the traditional CNN framework by integrating quantum computing principles to enhance feature extraction and classification. The input image is encoded into quantum states, as shown in Fig. 2, with each pixel or feature mapped to a qubit. This step prepares the data for quantum operations. The first layer in the QCNN applies unitary gates ( $U_1$ ) to these qubits, extracting essential quantum features [12]. These gates act as quantum equivalents of classical convolutional filters and operate in parallel, ensuring efficient feature extraction. The quantum pooling layer then reduces the complexity of the data by introducing entanglement between qubits using gates. This step allows the network to capture correlations between features while retaining only the most relevant information through quantum measurements. The refined quantum features are further processed by a second quantum convolutional layer, which applies additional unitary gates ( $U_2$ ) to identify more intricate relationships. A second

pooling layer reduces the feature map further, ensuring that only the critical features are retained.

The final stage of the QCNN architecture is the fully connected layer ( $F$ ), which integrates the quantum features into a classical representation for classification. This layer outputs the prediction, determining whether the input image is benign or malignant. By leveraging quantum properties such as superposition and entanglement, QCNN can efficiently handle high-dimensional data while capturing complex relationships. The combination of quantum operations and hierarchical learning makes QCNN a promising tool for medical image analysis, especially for challenging tasks like breast cancer classification.

#### IV. METHODOLOGY

The overall methodological workflow, shown in Fig.3, demonstrates the process of comparing CNN and QCNN models for breast cancer histopathological image classification. It begins with data acquisition from Kaggle, featuring labeled images of benign and malignant tumors. These images are preprocessed through resizing, normalization, and data augmentation to ensure consistency and robustness. The dataset is split into training, validation, and testing subsets. The training dataset is used to train both models, while the validation dataset fine-tunes hyperparameters such as batch size and epochs. The test dataset ensures unbiased evaluation on unseen data. During training, the CNN and QCNN models are optimized, with QCNN leveraging quantum-specific oper-

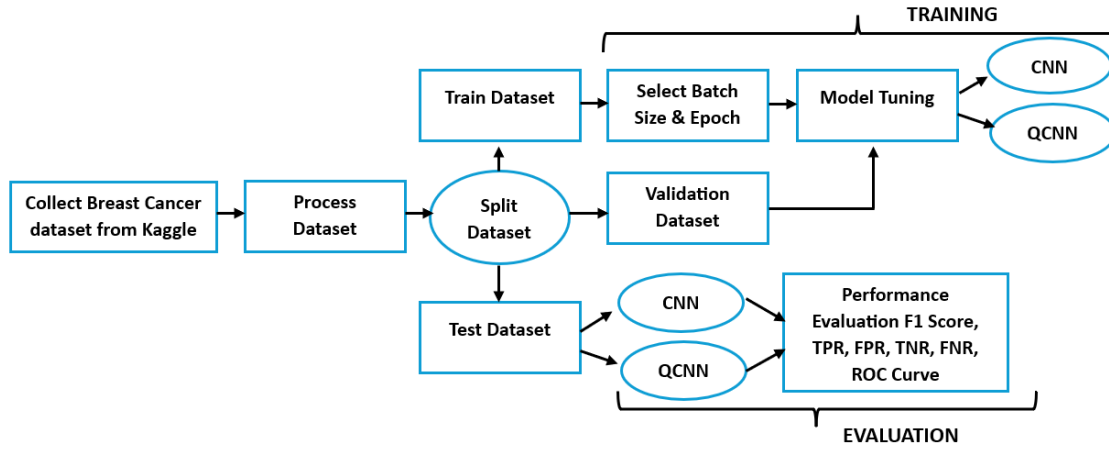


Fig. 3. Workflow for Comparing CNN and QCNN Models.

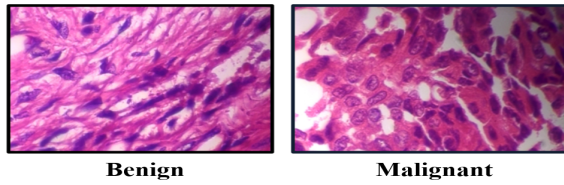


Fig. 4. Sample images used in dataset.

ations like quantum gates and circuits to improve performance. The evaluation phase assesses the models on the test dataset using metrics such as F1 score, true positive rate (TPR), false positive rate (FPR), and the ROC curve with its AUC. These metrics offer a detailed view of the models' accuracy, sensitivity, and classification performance. Further details about these metrics are provided in Section V: Evaluation Metrics. Finally, the outputs of CNN and QCNN are compared to identify their strengths and weaknesses in terms of accuracy. This structured workflow integrates traditional machine learning with quantum techniques, offering a robust framework for assessing the potential of quantum-based neural networks in medical imaging.

#### A. Data Acquisition

This study utilized data from Kaggle, a platform known for its diverse and comprehensive datasets, including widely recognized collections related to breast cancer. Specifically, we worked with the BreakHis Breast Cancer Histopathological Database [13]. The BreakHis dataset contains 9,109 microscopic images of breast tumor tissues collected from 82 patients. These images are categorized into two primary groups: benign tumors (2,480 samples) and malignant tumors (5,429 samples) and were captured at four magnification levels—40x, 100x, 200x, and 400x. For this research, we selected a subset of 550 benign and 770 malignant images at 400x magnification, which was chosen because it provides detailed cellular structures essential for distinguishing between benign and malignant tumors in binary classification tasks. To

ensure consistency and robustness, the images were resized to 100x100 pixels, preserving key features essential for accurate classification.

Kaggle's high-quality breast cancer image dataset, accompanied by detailed metadata and labels, is well-suited for studies involving CNN and QCNN models. Fig. 4, showcases histopathological images of breast tumor tissues from the dataset, highlighting both benign and malignant cases, which are crucial for evaluating model performance in binary classification tasks. Batch sizes (16, 32, and 64) and epochs (5, 10, 15, and 20) were employed to systematically evaluate model performance under varying conditions. Additionally, a set of 100 images was reserved for final testing to provide a robust and unbiased validation of the models' accuracy. This structured approach enabled the effective assessment of machine learning algorithms, specifically CNN and QCNN, in breast cancer classification.

#### B. Tools, Training Environment, and Hardware Integration

Our QCNN and CNN models were developed and tested in Google Colab, utilizing Google Cloud's GPU-A100 resources to accelerate the training process [14]. The integration with Keras allowed for seamless implementation of deep learning models, while the availability of GPUs facilitated faster training on large datasets [15]. However, Colab's dynamic resource allocation can cause fluctuations in performance. Selecting appropriate hardware, such as the GPU-A100, was essential in enhancing the performance of both CNN and QCNN models during training and testing [16]. To minimize these issues and ensure reliable outcomes, we adopted controlled resource conditions. Specifically, we standardized hardware usage by consistently using the GPU-A100 throughout all training and testing phases. Additionally, we maintained a consistent software environment by fixing the versions of essential libraries, such as TensorFlow, TensorFlow Quantum (TFQ), and Cirq, to avoid discrepancies arising from updates or compatibility issues. In the QCNN framework, we focused on enhancing quantum circuits, allocating qubits for accurate

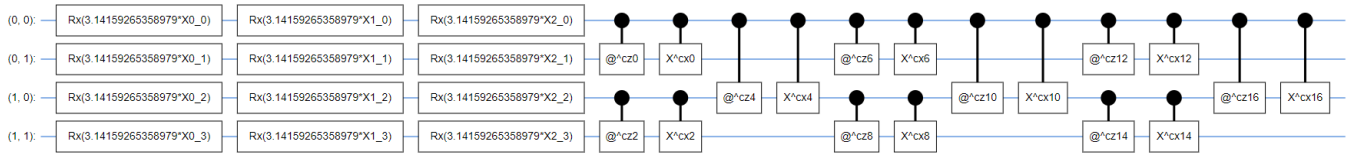


Fig. 5. QCNN Circuit Design.

feature representation, and tuning quantum gates to ensure operations perform accurately, with careful monitoring of performance across epochs and batch sizes. Reproducibility was an important part of our process. We repeated each experiment under the same conditions to reduce the impact of small variations. We also used consistent settings, such as batch size and number of epochs to ensure a fair comparison between the CNN and QCNN models.

## V. EVALUATION METRICS

We utilized various evaluation metrics to measure the performance of our model on unseen data, precisely in the context of classification tasks. This section briefly describes these metrics, which are essential for validating the practical utility of any binary classification model. The primary metrics in our evaluation are the True Positive Rate (TPR) and the False Positive Rate (FPR).

$$\text{True Positive Rate (TPR)} = \frac{TP}{TP + FN} \quad (1)$$

$$\text{False Positive Rate (FPR)} = \frac{FP}{FP + TN} \quad (2)$$

Another essential metric is the F1 score, which balances precision and recall.

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

Furthermore, the Receiver Operating Characteristic (ROC) curve, accompanied by the Area Under the Curve (ROC AUC), offers a graphical representation of the model's sensitivity-specificity trade-off. The ROC curve plots TPR against FPR across various decision thresholds, illustrating how the model's performance varies with different classification criteria. On the other hand, ROC AUC summarizes the model's overall discriminatory power, encapsulating its performance across all possible decision thresholds. A higher ROC AUC value indicates superior performance, distinguishing between the two classes and providing a comprehensive assessment of the model's effectiveness.

## VI. EXPERIMENTAL SETUP

For the CNN implementation, images were split into training and validation sets. Data augmentation techniques, such as transformations, were applied to enhance dataset diversity and improve model robustness. Additionally, pixel values were normalized to fall within the [0, 1] range. The CNN model was built using TensorFlow and Keras libraries. The CNN architecture featured two convolutional layers, each using ReLU activation and max-pooling layers to downsample feature maps. This was followed by a flattening layer and two fully connected (dense) layers, with the final output layer utilizing a sigmoid activation function for binary classification. The model was optimized using the Adam optimizer and binary cross-entropy loss function. Training was conducted with varying batch sizes and epochs, and performance was tracked through validation loss and accuracy metrics.

We created a custom quantum convolutional layer (QConv) for the QCNN using TensorFlow Quantum and Cirq to integrate quantum computing techniques into our model. This process required optimizing quantum circuits to allow for parallel processing, improving the allocation of qubits, and adjusting hardware parameters to enhance precision and overall performance. As shown in Fig. 5, the QConv layer operates on a grid of four qubits arranged in a 2x2 format, labeled by the coordinates (0, 0), (0, 1), (1, 0), and (1, 1). The circuit processes three input channels, corresponding to the input data features. The initial layer applies Rx rotations, parameterized by the input features, with each qubit controlled by parameters X00, X01, X02, and X03. Entanglement between qubits is achieved using Controlled-Z (CZ) and Controlled-X (CX) gates, applied sequentially at varying distances. These CZ and CX gates are governed by learnable parameters, such as cz0, cz2, cz4, cx0, cx2, and cx4, which are adjusted during training to optimize the quantum layer's performance. Following the quantum convolution, a sigmoid activation function is applied to introduce non-linearity into the model.

## VII. RESULTS AND DISCUSSION

This section presents the comparative analysis of QCNN and CNN performance in classifying breast cancer images, with a focus on accuracy and diagnostic reliability. The confusion matrix in Table I highlights the performance comparison between CNN and QCNN models in classifying breast cancer histopathological images. QCNN demonstrates superior accuracy, correctly identifying 49 actual positive cases with only 1 false negative, while CNN identifies 43 positives but misclassifies 7 as negative. For actual negative cases, QCNN

again outperforms CNN by correctly classifying 47 out of 50 negatives, with only 3 false positives, compared to CNN's 39 true negatives and 11 false positives. Overall, QCNN shows higher precision and sensitivity in detecting both positive and negative cases, reflecting better classification performance than CNN.

Table II presents the performance comparison between CNN and QCNN for batch 64 at epoch 10, using F1 Score and Accuracy as evaluation metrics. The QCNN model significantly outperforms CNN, particularly in handling class imbalances, which highlights its potential for high-precision and high-recall applications. With an F1 Score of 0.97 and an accuracy of 97%, QCNN demonstrates superior reliability in classifying both positive and negative cases compared to the CNN model, which shows an F1 Score of 0.82 and an accuracy of 82%.

The bar graph in Fig. 6 highlights how well CNN and QCNN models detect true positives (TP) across different batch sizes (16, 32, and 64). The QCNN model consistently achieves nearly 50 true positives at every batch size, demonstrating its strong ability to accurately identify positive cases, regardless of changes in batch size. In comparison, the CNN model struggles more with smaller batch sizes, detecting fewer true positives at batch sizes 16 and 32. While CNN shows improvement at batch size 64, it still doesn't match the consistent performance of QCNN. This shows that QCNN outperforms CNN across all batch sizes, offering higher accuracy and reliability. Overall, QCNN proves to be a robust and efficient model, especially when handling smaller batches, making it a more reliable choice for precise classification tasks.

This comparison of true positive rates across epochs provides insights into the performance of CNN and QCNN models during training, as depicted in Fig. 7. The bar graph highlights the differences in TP detection over various epochs (5, 10, 15, and 20). The QCNN model demonstrates superior accuracy, particularly at epoch 5, where it identifies more true positives than CNN. As training progresses, both models improve, with QCNN maintaining nearly 50 true positives from epoch 10 onward. While CNN also improves with more epochs, it consistently falls slightly behind QCNN. This analysis emphasizes QCNN's ability to achieve faster convergence and superior classification accuracy, making it highly effective for tasks requiring early-stage performance and reliability.

TABLE I  
TABLE I: CONFUSION MATRIX FOR CNN AND QCNN MODELS

|                        | Predicted Positive |             | Predicted Negative |             |
|------------------------|--------------------|-------------|--------------------|-------------|
|                        | <i>CNN</i>         | <i>QCNN</i> | <i>CNN</i>         | <i>QCNN</i> |
| <b>Actual Positive</b> | 43                 | 49          | 7                  | 1           |
| <b>Actual Negative</b> | 11                 | 3           | 39                 | 47          |

In Fig. 8, the bar graph compares the TN detection performance of CNN and QCNN across batch sizes (16, 32, 64). At batch size 16, QCNN significantly outperforms CNN, indicating better performance in smaller batches. As the batch size increases to 32, the gap narrows, with QCNN maintaining a slight advantage. By batch size 64, both models show nearly

TABLE II  
TABLE II: PERFORMANCE COMPARISON OF CNN AND QCNN

| Metric   | <i>CNN</i> | <i>QCNN</i> |
|----------|------------|-------------|
| F1 Score | 0.82       | 0.97        |
| Accuracy | 82%        | 97%         |

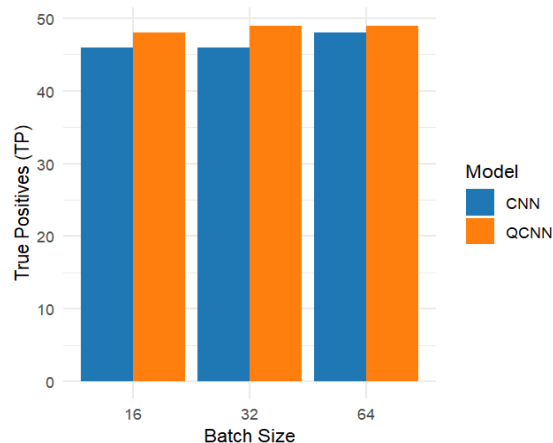


Fig. 6. True Positives for CNN and QCNN across different batch sizes.

identical accuracy. These results suggest that QCNN performs consistently well across all batch sizes, especially excelling in smaller ones, while CNN improves gradually as batch size increases.

The bar graph in Fig. 9 highlights the TN detection performance of CNN and QCNN models across epochs (5, 10, 15, 20). QCNN shows a clear advantage at epoch 5, detecting more true negatives, while the gap narrows at epochs 10 and 15 as both models improve. By epoch 20, their performance converges, with similar accuracy in TN detection. This emphasizes QCNN's faster convergence and CNN's steady improvement over time.

The ROC curve compares the classification performance between the CNN and QCNN models based on their TPR

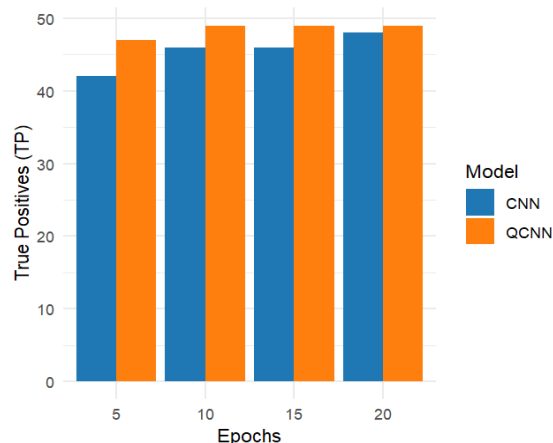


Fig. 7. True Positives for CNN and QCNN across different epochs.

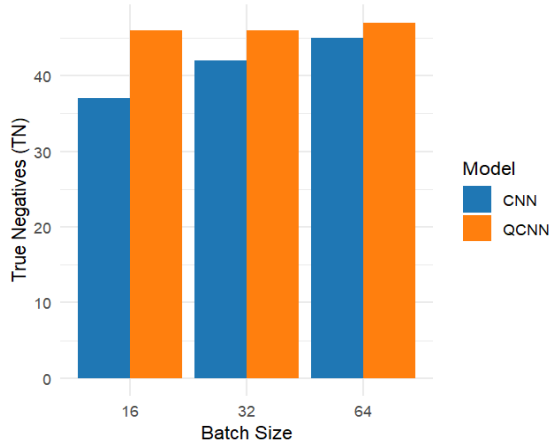


Fig. 8. True Negatives for CNN and QCNN across different batch sizes.

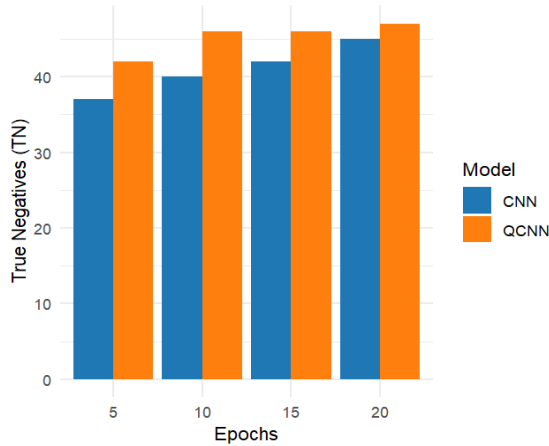


Fig. 9. True Negatives for CNN and QCNN across different epochs.

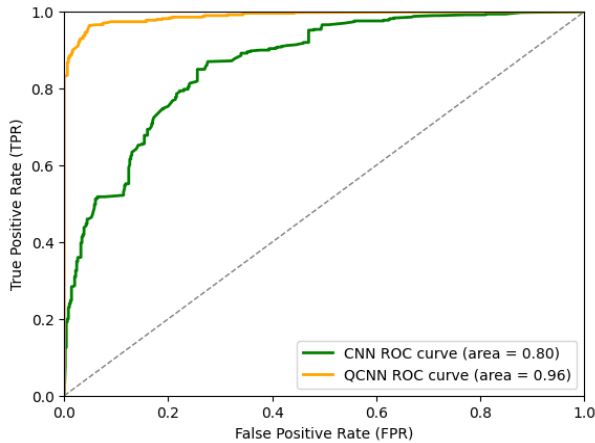


Fig. 10. ROC curve comparing CNN and QCNN models.

and FPR (Fig. 10). The QCNN model, with an area under the curve (AUC) of 0.96, exhibits superior performance by remaining close to the top-left corner, signifying its high accuracy in distinguishing between positive and negative cases. In contrast, the CNN model shows an AUC of 0.80, reflecting a solid performance but with more variability and a higher false positive rate, as evident from its curve's position. This difference indicates that QCNN is more effective in reducing misclassifications and achieving consistent precision. Overall, the graph emphasizes QCNN's enhanced ability to handle classification tasks with greater accuracy, making it a more reliable choice for distinguishing between classes than CNN.

## VIII. CONCLUSION

This research provides an in-depth analysis of the performance differences between CNN and QCNN, focusing on their application to binary classification of breast cancer histopathological images. By leveraging quantum principles, QCNN demonstrated superior accuracy and addressing the challenges of large datasets and complex features. These findings underscore QCNNs' potential to revolutionize medical image analysis, offering more accurate diagnostic tools that can significantly improve early detection and treatment planning for diseases like breast cancer.

Building on these results, future research should focus on optimizing quantum circuits for better efficiency and exploring the scalability of QCNN with larger, more diverse datasets. While this study focused on breast cancer classification, the methods and findings are broadly applicable. Expanding CNN and QCNN applications to multiclass classification tasks, drug discovery, and natural language processing can unlock further potential across diverse fields, validating their role in advancing machine learning and quantum computing.

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